DOCUMENT INFORMATION

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1. Introduction

1.1. Purpose of document

Technology documentation is a description of the tools, programming languages, and technologies used in a software project. It helps team members and stakeholders understand the technology used, supports development planning and strategy, and ensures consistency across the project.

1.2. Document objectives

The goal of a technology document is to provide a clear overview of the use of technology in a project in an effort to ensure the most efficient and effective use of the development team.

1.3. Intended audience

+ The technology stack document is intended for the following audiences:
+ Developers: To understand the technologies being used and adhere to the project standards.
+ Software Architects: To evaluate and verify the logic, scalability, and appropriateness of the chosen technologies.
+ Project Managers: To plan and estimate costs and timelines based on the selected technology stack.
+ New Team Members: To quickly grasp the project's technologies and processes.
+ Stakeholders: To have an overview of the project's development direction and scalability for future growth.

1.4. Acronyms and abbreviations

- Acronym/Abbreviation: EHMAS
- Meaning: Elderly Health Monitoring and Alert System

2. Main Decision to Choose Technologies

We have some key decisions to choose our technologies stack:

+ JavaScript is a widely used language that is well-supported in web development, while TypeScript, with its type-checking capabilities, enhances code quality and reduces errors during the development process.

+ ReactJS and React Native stands out for its ability to build flexible and interactive user interfaces through a component-based architecture, allowing for code reuse and speeding up development.
+ NodeJS with TypeScript enables the use of JavaScript on both the client and server sides, facilitating the development of a comprehensive application.
+ All of these technologies are not only modern and powerful but also backed by a large support community, ensuring that our project is developed safely and efficiently.

3. Programing Language

JavaScript, often abbreviated as JS, is a high-level, dynamic programming language that supports multiple programming paradigms. TypeScript, an extension of JavaScript, improves code stability and maintainability by adding static type checking. JavaScript's popularity makes it easy to find solutions on forums like Google or Stack Overflow.

4. Technologies of Web Applications

4.1. ReactJS

Home: https://reactjs.org/

ReactJS is a popular JavaScript library for building user interfaces. It is fast, efficient, and easy to extend. The large React community makes finding solutions and documentation easy, even for less experienced developers.

- Pros of React

- The community

+ React has a large community, making it easy for developers to find support.

Faster development

+ React helps shorten development time thanks to its component-based structure.

+ With React, the development time is considerably shorter.

4.2. NodeJS with TypeScript

NodeJS is a powerful JavaScript runtime, combined with TypeScript to improve code stability and maintainability. Using NodeJS, we can create flexible and scalable server-side applications, thanks to the power of Google V8 Engine and the ability to handle large numbers of concurrent requests.

4.3. Restful API

RESTful API is an architectural style for an application program interface (API) that uses HTTP requests to access and use data. We build APIs based on RESTful principles for easy data accessing and interacting between different levels of the system.

4.4. ExpressJS

Express is a NodeJS framework that simplifies server development, making it fast and flexible. It helps build RESTful APIs, making it an ideal choice for managing data and communicating between different components in the system.

5. Technologies Of Applications

5.1. React Native

Home: https://reactnative.dev/

React Native is a widely used JavaScript framework for building cross-platform mobile applications. It enables developers to create mobile apps for both iOS and Android using a single codebase, while still delivering near-native performance and smooth user experience.

- Pros of React Native
- Cross-platform development
  + Code can be reused across iOS and Android, reducing time and cost.
- High performance
  + Uses native components, ensuring fast rendering and smooth UI interactions.
- Fast development
  + Hot Reloading allows developers to see changes instantly.
- Large community
  + Rich ecosystem of plugins and community support improves productivity.

5.2. Mobile Application Using React Native

The mobile app built with React Native acts as the client interface for users, supporting essential business operations such as authentication, viewing services, placing orders, uploading files, and managing user profiles.

- Key Capabilities

+ Communicates with the backend through RESTful APIs.
+ Stores session tokens securely using local storage mechanisms.
+ Implements navigation via React Navigation for smooth user flows.
+ Supports optimized UI components for better usability on mobile devices.

5.3. Mobile Communication via RESTful API

RESTful API serves as the data bridge between the mobile app and backend services. The mobile application interacts with the server through standard HTTP methods such as GET, POST, PUT, and DELETE.

+ Pros of Mobile API Integration
+ Consistent data exchange
+ JSON format ensures lightweight and easy-to-parse responses.
+ Easy integration
+ The app can quickly fetch services, submit orders, and update information.
+ Security
+ Supports token-based authentication for secure data access.

5.4. ExpressJS for Mobile API Backend

ExpressJS provides an efficient and flexible backend layer that processes requests from the mobile application. It handles routing, validation, authentication, error handling, and communication with the database.

- Pros of ExpressJS for Mobile
- Fast server development

+ Minimal and flexible structure, easy to build and scale.

- Middleware support
  + Enhances security, processing, and request validation.
- Compatibility
  + Fully compatible with RESTful design, making it ideal for mobile apps.

6. Database

MySQL is a NoSQL database that is ideal for managing large amounts of data with flexible structures. MongoDB's document-based structure makes it easy to store and retrieve data, making it especially suitable for modern applications.

7. Development Stack

7.1. Visual Studio Code

Home page: http://code.visualstudio.com/

Visual Studio Code (Vscode) is a great open-source code editor developed by Microsoft it provides a rich and easy to use user interface as well as thousands of extensions that we can install to suit our needs.

7.2. Figma

Home: https://www.figma.com/

Figma is a powerful UI design and prototyping tool. With real-time collaboration capabilities, Figma allows team members to work in sync and easily share designs. This helps development teams understand and realize design elements in products quickly and accurately.

8. Version Control

8.1. Github

Home page: https://github.com

GitHub is a repository hosting service. Think of it as the "cloud" for code.

GitHub will host your source code projects in a variety of different programming languages and keep track of the various changes made to every iteration. It is able to do this by using git, a revision control system that runs in the command line interface.

Using GitHub has numerous benefits including easier collaboration with colleagues and peers, ability to look back on previous versions, and tons of easy integration options.

9. Deployment

9.1. Cloudinary

Home: https://cloudinary.com/

Cloudinary is a cloud-based media management service that supports uploading, storing, and optimizing images and files.

In this project, Cloudinary is used to store elderly profile photos, medical images, and other documents uploaded from the mobile application.

Cloudinary provides easy configuration and detailed documentation, helping the system handle media quickly and securely.

9.2. API Deployment

The EHMAS system deploys its backend API using a cloud server environment. The API is built with NodeJS and ExpressJS, providing the main communication layer between the mobile application and the database.

The cloud server allows stable API operation, making the system more reliable and easier to maintain.

Through this deployed API, the system can process health data, manage users, and integrate with Cloudinary for media handling.